

# MENTORA ACADEMY

EXCELLENCE IN EDUCATION

## Edexcel International A Level Chemistry (WCH14)

### UNIT 4 — MASTER TOPICAL PRACTICE PACK: 13A

#### CHEMICAL EQUILIBRIA (K<sub>c</sub>, K<sub>p</sub>, FACTORS & TOTAL ENTROPY RELATION)

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| <b>CANDIDATE NAME:</b>     | Usman                                                                                         |
| <b>SUBJECT &amp; CODE:</b> | Chemistry Unit 4 — Rates, Equilibria & Organic (WCH14)                                        |
| <b>TOPIC AREA:</b>         | Topic 13: CHEMICAL EQUILIBRIA                                                                 |
| <b>SUB-TOPIC:</b>          | 13A: Chemical Equilibria (K <sub>c</sub> , K <sub>p</sub> , Factors & Total Entropy Relation) |
| <b>TOTAL QUESTIONS:</b>    | 50 Authentic Past Paper Questions (Tier 1 & Tier 2)                                           |
| <b>TOTAL MARKS:</b>        | 229 Marks                                                                                     |
| <b>EXAMINER FAQs:</b>      | 10 Empirical Examiner Traps & Model Answers Included                                          |

#### HOW TO USE THIS A\* TOPICAL PACK:

1. Tier 1 (Q1–Q25): Mid-Easy to Moderate foundational questions (1–3 marks, MCQs, core calculations).
2. Tier 2 (Q26–Q50): Hard & A\* Challenge questions (4–6 marks, multi-step calculations, multi-spectral NMR, extended answers).
3. Review the complete worked mark scheme in Section C after attempting each tier.
4. Master the 10 Empirical Examiner FAQs in Section D to eliminate common candidate traps.

**SECTION A: TIER 1 PRACTICE QUESTIONS (FOUNDATIONAL TO MODERATE) — Q1 to Q25****Q1. 1. Equilibrium Constant Kc Expression & Units****[3 Marks]**  
**WCH14/01/Jan23/Q9**For the reversible reaction:  $\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \rightleftharpoons 2\text{NH}_3(\text{g})$ 

- (a) Write the expression for the equilibrium constant, Kc. [1]

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- (b) Deduce the units of Kc for this reaction. [2]

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**Q2. 2. Homogeneous vs Heterogeneous Equilibria Kc Rules****[4 Marks]**  
**WCH14/01/Oct22/Q10**Consider the heterogeneous equilibrium:  $\text{CaCO}_3(\text{s}) \rightleftharpoons \text{CaO}(\text{s}) + \text{CO}_2(\text{g})$ 

- (a) Explain why solid CaCO
- <sub>3</sub>
- and CaO are omitted from the Kc expression. [2]

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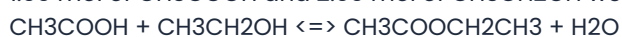
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- (b) Write the simplified Kc expression and state its units. [2]

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**Q3. 3. ICE Table Calculation for Kc****[5 Marks]**  
**WCH14/01/Jun22/Q11**1.00 mol of CH<sub>3</sub>COOH and 2.00 mol of CH<sub>3</sub>CH<sub>2</sub>OH were reacted in a 1.00 dm<sup>3</sup> vessel:

At equilibrium, 0.80 mol of ester was formed.

- (a) Calculate the equilibrium amounts of CH
- <sub>3</sub>
- COOH, CH
- <sub>3</sub>
- CH
- <sub>2</sub>
- OH, and H
- <sub>2</sub>
- O. [3]

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- (b) Calculate Kc for esterification. [2]

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**Q4. 4. Equilibrium Constant Kp Expression & Partial Pressures****[4 Marks]**  
**WCH14/01/Jan22/Q12**For  $2\text{SO}_2(\text{g}) + \text{O}_2(\text{g}) \rightleftharpoons 2\text{SO}_3(\text{g})$  at total pressure P<sub>total</sub>.

- (a) Define mole fraction x
- <sub>A</sub>
- and partial pressure p
- <sub>A</sub>
- . [2]

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(b) Write the expression for  $K_p$  and deduce its units. [2]

#### Q5. 5. Equilibrium Pressure Calculation for $K_p$

[5 Marks]  
WCH14/01/Oct21/Q11

In a mixture of  $N_2O_4$  and  $NO_2$  at 298 K, total pressure = 2.00 atm.

Mole fraction of  $N_2O_4$  = 0.40.

(a) Calculate the partial pressures of  $N_2O_4$  and  $NO_2$ . [2]

(b) Write  $K_p$  for  $N_2O_4(g) \rightleftharpoons 2NO_2(g)$  and calculate its value with units. [3]

#### Q6. 6. Effect of Temperature on $K_c$ and $K_p$ for Exothermic Reactions

[4 Marks]  
WCH14/01/Jun21/Q12

For  $N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$  ( $\Delta H = -92.2 \text{ kJ mol}^{-1}$ ).

(a) Explain why increasing temperature decreases the value of  $K_c$ . [3]

(b) Relate this to Le Chatelier's principle. [1]

#### Q7. 7. Effect of Temperature on $K$ for Endothermic Reactions

[3 Marks]  
WCH14/01/Jan21/Q13

For  $N_2O_4(g) \rightleftharpoons 2NO_2(g)$  ( $\Delta H = +57.2 \text{ kJ mol}^{-1}$ ).

(a) Predict and explain how increasing temperature affects  $K_p$ . [3]

**Q8. 8. Why Pressure and Concentration Changes Do NOT Change K****[ 4 Marks ]**  
**WCH14/01/Oct20/Q11**

For  $2\text{SO}_2(\text{g}) + \text{O}_2(\text{g}) \rightleftharpoons 2\text{SO}_3(\text{g})$ , total pressure is doubled.

- (a) Explain why doubling total pressure shifts equilibrium position to the right but leaves  $K_p$  unchanged. [4]

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**Q9. 9. Why Catalysts Do NOT Change Equilibrium Constant K****[ 3 Marks ]**  
**WCH14/01/Jun20/Q13**

A catalyst is added to an equilibrium mixture.

- (a) Explain why a catalyst increases the rate of achieving equilibrium but has zero effect on the value of  $K_c$  or  $K_p$ . [3]

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**Q10. 10. Derivation & Calculation of  $\Delta S_{\text{total}} = R \ln K$** **[ 4 Marks ]**  
**WCH14/01/Jan20/Q14**

At thermodynamic equilibrium, total entropy change relates to  $K$  by  $\Delta S_{\text{total}} = R \ln K$ .

- (a) Calculate  $\Delta S_{\text{total}}$  at 298 K for a reaction with  $K = 1.80 \text{ atm}$ . [2]

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- (b) Calculate  $K$  if  $\Delta S_{\text{total}} = +50.0 \text{ J K}^{-1} \text{ mol}^{-1}$ . [2]

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**Q11. 11. Degree of Dissociation Calculation for  $\text{PCl}_5$** **[ 5 Marks ]**  
**WCH14/01/Oct19/Q14**

$\text{PCl}_5(\text{g}) \rightleftharpoons \text{PCl}_3(\text{g}) + \text{Cl}_2(\text{g})$ . 1.00 mol of  $\text{PCl}_5$  was heated in a 10.0 dm<sup>3</sup> vessel. At equilibrium, 0.40 mol of  $\text{Cl}_2$  was present.

- (a) Calculate degree of dissociation  $\alpha$ . [1]

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- (b) Calculate  $K_c$  including units. [4]

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**Q12. 12. Esterification Equilibrium Calculation with Volume Cancellation****[ 4 Marks ]****WCH14/01/Jun19/Q15**

$\text{CH}_3\text{COOH} + \text{C}_2\text{H}_5\text{OH} \rightleftharpoons \text{CH}_3\text{COOC}_2\text{H}_5 + \text{H}_2\text{O}$ .  $K_c = 4.00$ .

(a) Explain why the container volume  $V$  cancels out in the  $K_c$  expression for this reaction. [2]

(b) Calculate moles of ester formed when 1.00 mol of acid and 1.00 mol of alcohol reach equilibrium. [2]

**Q13. 13. Kp Calculation for Methanol Synthesis****[ 5 Marks ]****WCH14/01/Jan19/Q16**

$\text{CO(g)} + 2\text{H}_2\text{(g)} \rightleftharpoons \text{CH}_3\text{OH(g)}$ . Equilibrium partial pressures:  $p_{\text{CO}} = 4.0 \text{ atm}$ ,  $p_{\text{H}_2} = 8.0 \text{ atm}$ ,  $p_{\text{CH}_3\text{OH}} = 2.0 \text{ atm}$ .

(a) Write  $K_p$  expression. [1]

(b) Calculate  $K_p$  and deduce units. [3]

(c) Predict effect of increasing total pressure on methanol yield. [1]

**Q14. 14. Relationship Between  $K_c$  and  $K_p$  via Ideal Gas Law****[ 4 Marks ]****WCH14/01/sample/Q14**

$K_p = K_c (R T)^{\Delta n}$  where  $\Delta n$  = change in gas moles.

(a) For  $\text{N}_2\text{(g)} + 3\text{H}_2\text{(g)} \rightleftharpoons 2\text{NH}_3\text{(g)}$ , state  $\Delta n$ . [1]

(b) Calculate  $K_p$  at 500 K if  $K_c = 0.060 \text{ dm}^6 \text{ mol}^{-2}$ . ( $R = 0.0821 \text{ dm}^3 \text{ atm K}^{-1} \text{ mol}^{-1}$ ) [3]

**Q15. 15. Heterogeneous Equilibrium Kp for Dehydration of Copper Sulfate****[3 Marks]****WCH14/01/sample/Q15**

(a) Write the Kp expression for this heterogeneous reaction. [1]

(b) If the equilibrium vapor pressure of water is 0.030 atm at 298 K, calculate Kp. [2]

**Q16. 16. ICE Table with Quadratic Equation Solution****[5 Marks]****WCH14/01/sample/Q16**

Initial:  $[\text{H}_2] = 1.00 \text{ M}$ ,  $[\text{I}_2] = 2.00 \text{ M}$ ,  $[\text{HI}] = 0.00 \text{ M}$ .

(a) Set up the Kc expression in terms of x (moles of H2 reacted). [2]

(b) Solve for x and calculate equilibrium concentrations of H2, I2, and HI. [3]

**Q17. 17. Temperature Dependence of K and Enthalpy Sign****[4 Marks]****WCH14/01/sample/Q17**

Kc for a reaction is  $1.0 \times 10^{-3}$  at 300 K and  $4.5 \times 10^{-2}$  at 400 K.

(a) Deduce whether the forward reaction is exothermic or endothermic. [2]

(b) Explain using van 't Hoff relationship  $\ln(K_2/K_1) = (-\Delta H / R)(1/T_2 - 1/T_1)$ . [2]

**Q18. 18. Calculation of Delta H from Two-Point K Data****[5 Marks]****WCH14/01/sample/Q18**

For  $\text{N}_2\text{O}_4 \rightleftharpoons 2\text{NO}_2$ : Kp = 0.141 atm at 298 K and Kp = 2.67 atm at 373 K.

(a) Calculate Delta H in kJ mol<sup>-1</sup> using  $\ln(K_2/K_1) = (\Delta H / R)(1/T_1 - 1/T_2)$ . [5]

[4 Marks]

**Q19. 19. Total Entropy at Equilibrium Boundary**

WCH14/01/sample/Q19

Explain why  $\Delta S_{\text{total}} = 0$  when a system reaches dynamic equilibrium.

- (a) Relate forward and reverse rates, entropy changes, and free energy at equilibrium. [4]

[4 Marks]

**Q20. 20. Le Chatelier's Principle vs Reaction Quotient Q**

WCH14/01/sample/Q20

For  $A + B \rightleftharpoons C + D$ ,  $K_c = 10.0$ .

A mixture currently has  $[A] = 0.5 \text{ M}$ ,  $[B] = 0.5 \text{ M}$ ,  $[C] = 2.0 \text{ M}$ ,  $[D] = 2.0 \text{ M}$ .

- (a) Calculate the reaction quotient  $Q_c = [C][D]/([A][B])$ . [2]

- (b) Compare  $Q_c$  with  $K_c$  and predict which direction the reaction will proceed to reach equilibrium. [2]

[4 Marks]

**Q21. 21. Effect of Inert Gas Addition on  $K_p$** 

WCH14/01/sample/Q21

Inert argon gas is added to  $N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$  at constant total volume.

- (a) Explain why adding an inert gas at constant volume has zero effect on the position of equilibrium or  $K_p$ . [4]

[4 Marks]

**Q22. 22. Effect of Inert Gas Addition at Constant Total Pressure**

WCH14/01/sample/Q22

Inert argon gas is added to  $N_2O_4(g) \rightleftharpoons 2NO_2(g)$  at constant total pressure.

- (a) Explain why adding argon at constant pressure causes the equilibrium to shift to the right. [4]

[ 5 Marks ]

**Q23. 23. Equilibrium Yield Optimization in Contact Process**

WCH14/01/Sample/Q23

Contact Process:  $2\text{SO}_2(\text{g}) + \text{O}_2(\text{g}) \rightleftharpoons 2\text{SO}_3(\text{g})$  ( $\Delta H = -197 \text{ kJ mol}^{-1}$ ).

(a) Explain the industrial choice of conditions: 1–2 atm pressure, 450 °C,  $\text{V}_2\text{O}_5$  catalyst. [5]

[ 4 Marks ]

**Q24. 24. Decomposition of Ammonium Chloride Equilibrium  
Kp**

WCH14/01/Sample/Q24

$\text{NH}_4\text{Cl}(\text{s}) \rightleftharpoons \text{NH}_3(\text{g}) + \text{HCl}(\text{g})$ . At equilibrium at 600 K, total pressure is 4.00 atm.

(a) Deduce partial pressures of  $\text{NH}_3$  and  $\text{HCl}$ . [2]

(b) Calculate  $K_p$  and give units. [2]

[ 5 Marks ]

**Q25. 25. Equilibrium Constant for Steam Reforming of  
Methane**

WCH14/01/Sample/Q25

$\text{CH}_4(\text{g}) + \text{H}_2\text{O}(\text{g}) \rightleftharpoons \text{CO}(\text{g}) + 3\text{H}_2(\text{g})$  ( $\Delta H = +206 \text{ kJ mol}^{-1}$ ).

(a) Write  $K_p$  expression. [1]

(b) Explain why high temperature and low pressure maximize equilibrium yield of  $\text{H}_2$ . [4]

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**SECTION B: TIER 2 A\* CHALLENGE QUESTIONS (HARD & EXTENDED) — Q26 to Q50****Q26. 26. A\* Challenge: Multi-Step ICE Table with Partial Pressures****[5 Marks]**  
**WCH14/01/Hard/Q26**

Equilibrium:  $A(g) + 2B(g) \rightleftharpoons C(g) + D(g)$ .

Initial: 2.0 mol A, 3.0 mol B in a container at 10.0 atm total pressure.

At equilibrium, 1.0 mol of C is formed.

- (a) Calculate equilibrium moles of A, B, C, D and total gas moles. [3]

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- (b) Calculate partial pressures and  $K_p$ . [2]

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**Q27. 27. A\* Challenge: Thermodynamic Proof of Le Chatelier's Principle****[5 Marks]**  
**WCH14/01/Hard/Q27**

Use  $\Delta S_{\text{total}} = \Delta S_{\text{system}} - \Delta H / T$  to prove that increasing temperature MUST decrease  $K$  for exothermic reactions.

- (a) Write the expression for  $\ln K$  as a function of  $1/T$ . [2]

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- (b) Differentiate  $\ln K$  w.r.t  $1/T$  (van 't Hoff equation  $d(\ln K)/d(1/T) = -\Delta H / R$ ) and explain the slope. [3]

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**Q28. 28. A\* Challenge: Calculation of Equilibrium Composition from  $K_c$** **[5 Marks]**  
**WCH14/01/Hard/Q28**

$N_2O_4(g) \rightleftharpoons 2NO_2(g)$ .  $K_c = 0.36 \text{ mol dm}^{-3}$  at 373 K.

If 1.00 mol of pure  $N_2O_4$  is placed in a 2.00 dm<sup>3</sup> flask.

- (a) Calculate the equilibrium concentration of  $NO_2$  and  $N_2O_4$ . [5]

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**Q29. 29. A\* Challenge: Simultaneous Chemical Equilibria Analysis****[5 Marks]**  
**WCH14/01/Hard/Q29**

Consider two coupled reactions:

Reaction 1:  $A \rightleftharpoons B$  ( $K_1 = 100$ )

Reaction 2:  $B \rightleftharpoons C$  ( $K_2 = 50$ )

(a) Deduce the equilibrium constant  $K_3$  for overall reaction  $A \rightleftharpoons C$ . [2]

(b) If starting with 1.0 M A, calculate the equilibrium concentrations of A, B, and C. [3]

**Q30. 30. A\* Challenge: Total Pressure Effect on Dissociation Fraction****[5 Marks]**  
**WCH14/01/Hard/Q30**

Show that for  $A(g) \rightleftharpoons 2B(g)$ , the degree of dissociation  $\alpha$  relates to total pressure  $P$  by  $\alpha = \sqrt{K_p / (K_p + 4P)}$ .

(a) Derive this formula starting from initial 1 mole of A. [5]

**Q31. 31. A\* Challenge: Temperature Dependence of  $K_p$  and Enthalpy Profile****[5 Marks]**  
**WCH14/01/Hard/Q31**

The equilibrium constant  $K_p$  for  $2NO_2(g) \rightleftharpoons N_2O_4(g)$  is 6.82 atm<sup>-1</sup> at 298 K and 0.144 atm<sup>-1</sup> at 373 K.

(a) Calculate  $\Delta H^\circ$  for dimerization of  $NO_2$ . [4]

(b) State whether dimerization is exothermic or endothermic. [1]

**Q32. 32. A\* Challenge: Equilibrium Constant Units in Terms of SI Units****[ 4 Marks ]**  
**WCH14/01/Hard/Q32**

For  $\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \rightleftharpoons 2\text{NH}_3(\text{g})$ ,  $K_p$  is expressed in  $\text{Pa}^{-2}$ .

- (a) Convert  $K_p = 3.56 \times 10^{-5} \text{ atm}^{-2}$  into  $\text{Pa}^{-2}$  ( $1 \text{ atm} = 101325 \text{ Pa}$ ). [2]

- (b) Express Pa in SI base units (kg, m, s). [2]

**Q33. 33. A\* Challenge: Gas Phase Decomposition of HI Kinetic vs Equilibrium Control****[ 5 Marks ]**  
**WCH14/01/Hard/Q33**

$2\text{HI}(\text{g}) \rightleftharpoons \text{H}_2(\text{g}) + \text{I}_2(\text{g})$  ( $\Delta H = +12.5 \text{ kJ mol}^{-1}$ ).  $K_c = 0.020$  at  $700 \text{ K}$ .

- (a) Calculate the percentage of HI decomposed at equilibrium. [3]

- (b) Explain why increasing temperature increases both rate of reaching equilibrium and equilibrium yield of  $\text{H}_2$ . [2]

**Q34. 34. A\* Challenge: Water Gas Shift Reaction Equilibrium****[ 5 Marks ]**  
**WCH14/01/Hard/Q34**

$\text{CO}(\text{g}) + \text{H}_2\text{O}(\text{g}) \rightleftharpoons \text{CO}_2(\text{g}) + \text{H}_2(\text{g})$  ( $\Delta H = -41.2 \text{ kJ mol}^{-1}$ ).  $K_c = 1.00$  at  $1100 \text{ K}$ .

- (a) If starting with  $1.00 \text{ mol CO}$  and  $2.00 \text{ mol H}_2\text{O}$ , calculate equilibrium composition. [3]

- (b) Explain why pressure has no effect on equilibrium composition. [2]

**Q35. 35. A\* Challenge: Relate  $K_p$  to Gibbs Free Energy Change  $\Delta G^\circ$** **[ 5 Marks ]**  
**WCH14/01/Hard/Q35**

$\Delta G^\circ = -RT \ln K_p$ .

- (a) Calculate  $\Delta G^\circ$  at  $298 \text{ K}$  for  $\text{N}_2\text{O}_4 \rightleftharpoons 2\text{NO}_2$  where  $K_p = 0.141 \text{ atm}$ . [3]

(b) State what a positive  $\Delta G^\circ$  indicates about standard state feasibility. [2]

**Q36. 36. A\* Challenge: Dissociation of Dinitrogen Tetroxide at Variable Volumes**

[5 Marks]

WCH14/01/Hard/Q36

$\text{N}_2\text{O}_4(\text{g}) \rightleftharpoons 2\text{NO}_2(\text{g})$ .  $K_c = 0.0046 \text{ mol dm}^{-3}$  at 298 K.

(a) Calculate degree of dissociation  $\alpha$  when 1.00 mol  $\text{N}_2\text{O}_4$  is in 10.0  $\text{dm}^3$  vs 100.0  $\text{dm}^3$ . [4]

(b) Explain Ostwald's dilution effect on gaseous dissociation. [1]

**Q37. 37. A\* Challenge: Synthesis of Phosgene  $\text{COCl}_2$  Equilibrium  $K_p$**

[5 Marks]

WCH14/01/Hard/Q37

$\text{CO}(\text{g}) + \text{Cl}_2(\text{g}) \rightleftharpoons \text{COCl}_2(\text{g})$ .  $K_p = 15.0 \text{ atm}^{-1}$  at 373 K.

Equilibrium partial pressure of  $\text{COCl}_2$  is 3.00 atm, and  $p_{\text{CO}} = p_{\text{Cl}_2}$ .

(a) Calculate the partial pressure of CO at equilibrium. [3]

(b) Calculate total pressure of the equilibrium mixture. [2]

**Q38. 38. A\* Challenge: Homogeneous Gas Phase Equilibrium Calculation for  $\text{SO}_3$  Formation**

[5 Marks]

WCH14/01/Hard/Q38

$2\text{SO}_2(\text{g}) + \text{O}_2(\text{g}) \rightleftharpoons 2\text{SO}_3(\text{g})$ .  $K_c = 280 \text{ dm}^3 \text{ mol}^{-1}$  at 1000 K.

In a 2.00  $\text{dm}^3$  vessel, equilibrium mixture contains 0.40 mol  $\text{SO}_2$  and 0.20 mol  $\text{O}_2$ .

(a) Calculate the concentration of  $\text{SO}_3$  at equilibrium. [3]

(b) Calculate total moles of SO<sub>3</sub> present in the 2.00 dm<sup>3</sup> vessel. [2]

**Q39. 39. A\* Challenge: Ammonia Decomposition K<sub>p</sub> and Partial Pressure Ratios**

[5 Marks]  
WCH14/01/Hard/Q39

$2\text{NH}_3(\text{g}) \rightleftharpoons \text{N}_2(\text{g}) + 3\text{H}_2(\text{g})$ . Pure NH<sub>3</sub> is introduced at initial pressure P<sub>0</sub>.

(a) Express K<sub>p</sub> in terms of equilibrium partial pressure p<sub>NH<sub>3</sub></sub> and mole fraction ratio. [3]

(b) If 20% of NH<sub>3</sub> decomposes at total pressure 10.0 atm, calculate K<sub>p</sub>. [2]

**Q40. 40. A\* Challenge: Le Chatelier Pressure Paradox in Gas Phase Reactions**

[5 Marks]  
WCH14/01/Hard/Q40

For  $\text{N}_2\text{O}_4(\text{g}) \rightleftharpoons 2\text{NO}_2(\text{g})$ , increasing pressure increases [NO<sub>2</sub>] in mol dm<sup>-3</sup> but decreases mole fraction of NO<sub>2</sub>.

(a) Explain this apparent paradox using Boyle's law and equilibrium shift. [5]

**Q41. 41. A\* Challenge: Calculation of K<sub>c</sub> for Ester Hydrolysis with Added Acid Catalyst**

[5 Marks]  
WCH14/01/Hard/Q41

1.00 mol ethyl ethanoate + 1.00 mol water + 0.10 mol HCl(aq) catalyst (containing water).

Equilibrium mixture titrated: 0.40 mol ester remains at equilibrium.

(a) Account for water contributed by HCl solution in the ICE table. [2]

(b) Calculate K<sub>c</sub>. [3]

**Q42. 42. A\* Challenge: Equilibrium Constant Kp of Steam Methane Reforming****[5 Marks]**  
**WCH14/01/Hard/Q42**

$\text{CH}_4(\text{g}) + \text{H}_2\text{O}(\text{g}) \rightleftharpoons \text{CO}(\text{g}) + 3\text{H}_2(\text{g})$ . At 1000 K, total pressure = 5.00 atm,  $p_{\text{CH}_4} = 0.50$  atm,  $p_{\text{H}_2\text{O}} = 0.50$  atm.

(a) Calculate  $p_{\text{CO}}$  and  $p_{\text{H}_2}$  given  $p_{\text{H}_2} = 3 p_{\text{CO}}$ . [2]

(b) Calculate  $K_p$ . [3]

**Q43. 43. A\* Challenge: Temperature Dependence of Solvation Equilibria****[5 Marks]**  
**WCH14/01/Hard/Q43**

Gas solubility equilibrium:  $\text{O}_2(\text{g}) \rightleftharpoons \text{O}_2(\text{aq})$  ( $\Delta H = -12.0 \text{ kJ mol}^{-1}$ ).

(a) Explain using thermodynamics why dissolved oxygen concentration in water decreases as water temperature rises. [3]

(b) Discuss environmental implications for aquatic life in thermal pollution. [2]

**Q44. 44. A\* Challenge: Multi-Gas Equilibrium Partitioning in Closed Container****[5 Marks]**  
**WCH14/01/Hard/Q44**

$\text{N}_2(\text{g}) + \text{O}_2(\text{g}) \rightleftharpoons 2\text{NO}(\text{g})$  ( $K_c = 1.0 \times 10^{-5}$  at 1500 K).

Air mixture initially: 0.80 M  $\text{N}_2$ , 0.20 M  $\text{O}_2$ .

(a) Calculate equilibrium concentration of NO. [3]

(b) Why can the approximation  $(0.80 - x) \sim 0.80$  be used? [2]

**Q45. 45. A\* Challenge: Van 't Hoff Isochore Derivation & Graph Analysis**

[5 Marks]  
WCH14/01/Hard/Q45

A plot of  $\ln K$  against  $1/T$  for a reversible reaction yields a straight line with gradient  $+4500 \text{ K}$  and  $y$ -intercept  $-12.5$ .

(a) Calculate  $\Delta H^\circ$  in  $\text{kJ mol}^{-1}$ . [2]

(b) Calculate  $\Delta S_{\text{system}}^\circ$  in  $\text{J K}^{-1} \text{ mol}^{-1}$ . [3]

**Q46. 46. A\* Challenge: Gas Phase Equilibrium Dissociation of  $\text{N}_2\text{O}_4$  at High Altitude**

[5 Marks]  
WCH14/01/Hard/Q46

At sea level ( $P = 1.0 \text{ atm}$ ),  $\text{N}_2\text{O}_4$  is 20% dissociated. At high altitude ( $P = 0.50 \text{ atm}$ ).

(a) Calculate  $K_p$  at  $298 \text{ K}$ . [2]

(b) Calculate degree of dissociation  $\alpha$  at  $0.50 \text{ atm}$ . [3]

**Q47. 47. A\* Challenge: Industrial Synthesis of Hydrogen Cyanide Equilibrium**

[5 Marks]  
WCH14/01/Hard/Q47

Andrussow process:  $\text{CH}_4(\text{g}) + \text{NH}_3(\text{g}) + 1.5 \text{ O}_2(\text{g}) \rightarrow \text{HCN}(\text{g}) + 3\text{H}_2\text{O}(\text{g})$  ( $\Delta H = -473 \text{ kJ mol}^{-1}$ ).

(a) Explain how temperature and pressure affect equilibrium yield vs rate. [3]

(b) Why is Pt-Rh gauze catalyst used at  $1100^\circ\text{C}$ ? [2]

**Q48. 48. A\* Challenge: Equilibrium Constant of Carbonate Buffer System****[5 Marks]****WCH14/01/Hard/Q48**

In ocean chemistry:  $\text{CO}_2(\text{aq}) + \text{H}_2\text{O}(\text{l}) \rightleftharpoons \text{H}_2\text{CO}_3(\text{aq}) \rightleftharpoons \text{H}^+(\text{aq}) + \text{HCO}_3^-(\text{aq})$ .

(a) Explain how increasing atmospheric  $\text{CO}_2$  partial pressure affects ocean pH (ocean acidification). [3]

(b) Relate this to carbonate ion dissolution  $\text{CaCO}_3(\text{s}) + \text{H}^+(\text{aq}) \rightarrow \text{Ca}^{2+}(\text{aq}) + \text{HCO}_3^-(\text{aq})$ . [2]

**Q49. 49. A\* Challenge: Equilibrium Analysis of Dehydration of Hydrated Cobalt Chloride****[5 Marks]****WCH14/01/Hard/Q49**

$\text{CoCl}_2 \cdot 6\text{H}_2\text{O}(\text{s})$  (pink)  $\rightleftharpoons \text{CoCl}_2(\text{s})$  (blue) +  $6\text{H}_2\text{O}(\text{g})$  ( $\Delta H = +350 \text{ kJ mol}^{-1}$ ).

(a) Explain why cobalt chloride paper turns blue in dry air and pink in humid air. [3]

(b) Write  $K_p = (p_{\text{H}_2\text{O}})^6$ . [2]

**Q50. 50. A\* Challenge: Complete Equilibria & Thermodynamics Master Synthesis****[6 Marks]****WCH14/01/Hard/Q50**

Synthesis of phosgene:  $\text{CO}(\text{g}) + \text{Cl}_2(\text{g}) \rightleftharpoons \text{COCl}_2(\text{g})$

$\Delta H^\circ = -108 \text{ kJ mol}^{-1}$ ,  $\Delta S_{\text{system}}^\circ = -132 \text{ J K}^{-1} \text{ mol}^{-1}$ .

(a) Calculate  $\Delta S_{\text{total}}$  and  $K_p$  at 298 K. [4]

(b) Calculate the temperature above which phosgene spontaneously dissociates into CO and  $\text{Cl}_2$ . [2]

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## SECTION C: WORKED MARK SCHEME (Q1–Q50) — 13A

| Q   | MARKING SCHEME & ACCEPTABLE ANSWERS                                                                                                                                                                                                                                                                                                                                                                                                                                 | MARKS |
|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| Q1  | Q1: 1. Equilibrium Constant $K_c$ Expression & Units<br>1. (a) $K_c = [\text{NH}_3]^2 / ([\text{N}_2][\text{H}_2]^3)$ (1).<br>1. (b) Units = $(\text{mol dm}^{-3})^2 / ((\text{mol dm}^{-3})(\text{mol dm}^{-3})^3) = 1 / (\text{mol dm}^{-3})^2 = \text{dm}^6 \text{ mol}^{-2}$ (2).                                                                                                                                                                               | [3]   |
| Q2  | Q2: 2. Homogeneous vs Heterogeneous Equilibria $K_c$ Rules<br>2. (a) Concentrations of pure solids are constant and incorporated into the value of $K_c$ (2).<br>2. (b) $K_c = [\text{CO}_2(\text{g})]$ (1); units = $\text{mol dm}^{-3}$ (1).                                                                                                                                                                                                                      | [4]   |
| Q3  | Q3: 3. ICE Table Calculation for $K_c$<br>3. (a) $\text{CH}_3\text{COOH eq} = 1.00 - 0.80 = 0.20 \text{ mol}$ (1). $\text{CH}_3\text{CH}_2\text{OH eq} = 2.00 - 0.80 = 1.20 \text{ mol}$ (1). $\text{H}_2\text{O eq} = 0.80 \text{ mol}$ (1).<br>3. (b) $K_c = (0.80 \times 0.80) / (0.20 \times 1.20) = 0.64 / 0.24 = 2.67$ (dimensionless) (2).                                                                                                                   | [5]   |
| Q4  | Q4: 4. Equilibrium Constant $K_p$ Expression & Partial Pressures<br>4. (a) Mole fraction $x_A = \text{moles A} / \text{total gas moles}$ (1). Partial pressure $p_A = x_A \times P_{\text{total}}$ (1).<br>4. (b) $K_p = (p_{\text{SO}_3})^2 / ((p_{\text{SO}_2})^2 (p_{\text{O}_2}))$ (1); units = $\text{atm}^{-1}$ or $\text{kPa}^{-1}$ or $\text{Pa}^{-1}$ (1).                                                                                                 | [4]   |
| Q5  | Q5: 5. Equilibrium Pressure Calculation for $K_p$<br>5. (a) $p_{\text{N}_2\text{O}_4} = 0.40 \times 2.00 = 0.80 \text{ atm}$ (1). $x_{\text{NO}_2} = 1.0 - 0.40 = 0.60 \Rightarrow p_{\text{NO}_2} = 0.60 \times 2.00 = 1.20 \text{ atm}$ (1).<br>5. (b) $K_p = (p_{\text{NO}_2})^2 / p_{\text{N}_2\text{O}_4} = (1.20)^2 / 0.80 = 1.44 / 0.80 = 1.80 \text{ atm}$ (3).                                                                                             | [5]   |
| Q6  | Q6: 6. Effect of Temperature on $K_c$ and $K_p$ for Exothermic Reactions<br>6. (a) Increasing temperature favours the endothermic reverse reaction to absorb heat (1). Concentration of reactants increases and products decreases (1). Since $K_c = [\text{NH}_3]^2 / ([\text{N}_2][\text{H}_2]^3)$ , $K_c$ decreases (1).<br>6. (b) Le Chatelier's principle states equilibrium shifts to oppose temperature increase (1).                                        | [4]   |
| Q7  | Q7: 7. Effect of Temperature on $K$ for Endothermic Reactions<br>7. (a) Forward reaction is endothermic ( $\Delta H > 0$ ) (1). Increasing temperature shifts equilibrium to the right to absorb heat (1). Partial pressure of $\text{NO}_2$ increases relative to $\text{N}_2\text{O}_4$ , so $K_p$ increases (1).                                                                                                                                                 | [3]   |
| Q8  | Q8: 8. Why Pressure and Concentration Changes Do NOT Change $K$<br>8. (a) Doubling pressure initially doubles all partial pressures (1). The denominator $(p_{\text{SO}_2})^2(p_{\text{O}_2})$ increases by $2^3 = 8x$ , while numerator $(p_{\text{SO}_3})^2$ increases by $2^2 = 4x$ (1). Quotient $Q_p < K_p$ (1). System responds by converting $2\text{SO}_2 + \text{O}_2$ into $2\text{SO}_3$ until partial pressure ratio restores exact value of $K_p$ (1). | [4]   |
| Q9  | Q9: 9. Why Catalysts Do NOT Change Equilibrium Constant $K$<br>9. (a) Catalyst speeds up both forward and reverse reaction rates by the exact same factor (1). Lowers activation energy $E_a$ for both directions equally (1). Ratio of rate constants $k_{\text{forward}} / k_{\text{reverse}} = K$ remains constant (1).                                                                                                                                          | [3]   |
| Q10 | Q10: 10. Derivation & Calculation of $\Delta S_{\text{total}} = R \ln K$<br>10. (a) $\Delta S_{\text{total}} = 8.31 \times \ln(1.80) = 8.31 \times 0.5878 = +4.88 \text{ J K}^{-1} \text{ mol}^{-1}$ (2).<br>10. (b) $\ln K = 50.0 / 8.31 = 6.0169 \Rightarrow K = e^{6.0169} = 410.3$ (2).                                                                                                                                                                         | [4]   |
| Q11 | Q11: 11. Degree of Dissociation Calculation for $\text{PCl}_5$<br>11. (a) $\alpha = 0.40 / 1.00 = 0.40$ (40%) (1).<br>11. (b) $[\text{PCl}_5]_{\text{eq}} = 0.60/10 = 0.060 \text{ M}$ ; $[\text{PCl}_3]_{\text{eq}} = 0.40/10 = 0.040 \text{ M}$ ; $[\text{Cl}_2]_{\text{eq}} = 0.040 \text{ M}$ (2). $K_c = (0.040 \times 0.040) / 0.060 = 0.00267 \text{ mol dm}^{-3}$ (2).                                                                                      | [5]   |
| Q12 | Q12: 12. Esterification Equilibrium Calculation with Volume Cancellation<br>12. (a) Equal number of moles of reactants (2 mol) and products (2 mol), so volume $V$ in numerator and denominator cancels completely (2).<br>12. (b) $K_c = x^2 / (1-x)^2 = 4.00 \Rightarrow x / (1-x) = 2.00 \Rightarrow x = 2 - 2x \Rightarrow 3x = 2 \Rightarrow x = 0.67 \text{ mol}$ (2).                                                                                        | [4]   |

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| <b>Q13</b> | Q13: 13. Kp Calculation for Methanol Synthesis<br>13. (a) $K_p = p_{\text{CH}_3\text{OH}} / (p_{\text{CO}} \times (p_{\text{H}_2})^2)$ (1).<br>13. (b) $K_p = 2.0 / (4.0 \times 64.0) = 2.0 / 256.0 = 7.81 \times 10^{-3} \text{ atm}^{-2}$ (3).<br>13. (c) Yield increases (3 gas moles form 1 gas mole) (1).                                                                                                                                                       | <b>[5]</b> |
| <b>Q14</b> | Q14: 14. Relationship Between Kc and Kp via Ideal Gas Law<br>14. (a) $\Delta n = 2 - 4 = -2$ (1).<br>14. (b) $K_p = K_c (RT)^{\Delta n} = 0.060 \times (0.0821 \times 500)^{-2} = 0.060 \times (41.05)^{-2} = 0.060 / 1685.1 = 3.56 \times 10^{-5} \text{ atm}^{-2}$ (3).                                                                                                                                                                                            | <b>[4]</b> |
| <b>Q15</b> | Q15: 15. Heterogeneous Equilibrium Kp for Dehydration of Copper Sulfate<br>15. (a) $K_p = (p_{\text{H}_2\text{O}})^5$ (1).<br>15. (b) $K_p = (0.030)^5 = 2.43 \times 10^{-8} \text{ atm}^5$ (2).                                                                                                                                                                                                                                                                     | <b>[3]</b> |
| <b>Q16</b> | Q16: 16. ICE Table with Quadratic Equation Solution<br>16. (a) $K_c = (2x)^2 / ((1-x)(2-x)) = 54.0 \Rightarrow 4x^2 = 54.0(2 - 3x + x^2) \Rightarrow 50.0x^2 - 162.0x + 108.0 = 0$ (2).<br>16. (b) $x = 0.93 \text{ mol dm}^{-3}$ (3). $[\text{H}_2] = 0.07 \text{ M}$ , $[\text{I}_2] = 1.07 \text{ M}$ , $[\text{HI}] = 1.86 \text{ M}$ .                                                                                                                          | <b>[5]</b> |
| <b>Q17</b> | Q17: 17. Temperature Dependence of K and Enthalpy Sign<br>17. (a) Endothermic ( $\Delta H > 0$ ) because Kc increases as temperature increases (2).<br>17. (b) $\ln(K_2/K_1) > 0$ and $(1/T_2 - 1/T_1) < 0$ , so $-\Delta H / R$ must be negative $\Rightarrow \Delta H > 0$ (2).                                                                                                                                                                                    | <b>[4]</b> |
| <b>Q18</b> | Q18: 18. Calculation of Delta H from Two-Point K Data<br>18. (a) $\ln(2.67 / 0.141) = \ln(18.936) = 2.941$ (1). $(1/298 - 1/373) = 6.749 \times 10^{-4} \text{ K}^{-1}$ (1). $\Delta H / R = 2.941 / 6.749 \times 10^{-4} = 4358 \text{ K}$ (2). $\Delta H = 4358 \times 8.31 = 36215 \text{ J mol}^{-1} = +36.2 \text{ kJ mol}^{-1}$ (1).                                                                                                                           | <b>[5]</b> |
| <b>Q19</b> | Q19: 19. Total Entropy at Equilibrium Boundary<br>19. (a) At dynamic equilibrium, rate of forward reaction equals rate of reverse reaction (1). Net concentration of all species remains constant (1). Universe entropy is maximized for the system composition, so $\Delta S_{\text{total}} = 0$ (1). $\Delta G = -T \Delta S_{\text{total}} = 0$ (1).                                                                                                              | <b>[4]</b> |
| <b>Q20</b> | Q20: 20. Le Chatelier's Principle vs Reaction Quotient Q<br>20. (a) $Q_c = (2.0 \times 2.0) / (0.5 \times 0.5) = 4.0 / 0.25 = 16.0$ (2).<br>20. (b) $Q_c (16.0) > K_c (10.0)$ (1). Reaction will shift to the left (reverse direction) to consume C and D and form more A and B until $Q_c = K_c$ (1).                                                                                                                                                               | <b>[4]</b> |
| <b>Q21</b> | Q21: 21. Effect of Inert Gas Addition on Kp<br>21. (a) Adding argon increases total pressure but total volume is constant (1). Partial pressures of N <sub>2</sub> , H <sub>2</sub> , and NH <sub>3</sub> remain completely unchanged (2). Therefore Kp and equilibrium position remain unchanged (1).                                                                                                                                                               | <b>[4]</b> |
| <b>Q22</b> | Q22: 22. Effect of Inert Gas Addition at Constant Total Pressure<br>22. (a) Adding argon increases total gas volume to keep total pressure constant (1). Mole fractions and partial pressures of reactant N <sub>2</sub> O <sub>4</sub> and product NO <sub>2</sub> both decrease (1). Since $K_p = (p_{\text{NO}_2})^2 / p_{\text{N}_2\text{O}_4}$ , denominator decreases less than numerator (1). System shifts right (towards more gas moles) to restore Kp (1). | <b>[4]</b> |
| <b>Q23</b> | Q23: 23. Equilibrium Yield Optimization in Contact Process<br>23. (a) Moderate pressure (1–2 atm) gives >99% yield without high cost of high-pressure equipment (2). 450 °C provides acceptable rate while keeping exothermic Kp high enough for high yield (2). V <sub>2</sub> O <sub>5</sub> catalyst speeds up rate without affecting yield (1).                                                                                                                  | <b>[5]</b> |
| <b>Q24</b> | Q24: 24. Decomposition of Ammonium Chloride Equilibrium Kp<br>24. (a) Equimolar gas formation $\Rightarrow p_{\text{NH}_3} = p_{\text{HCl}} = 2.00 \text{ atm}$ (2).<br>24. (b) $K_p = p_{\text{NH}_3} \times p_{\text{HCl}} = 2.00 \times 2.00 = 4.00 \text{ atm}^2$ (2).                                                                                                                                                                                           | <b>[4]</b> |
| <b>Q25</b> | Q25: 25. Equilibrium Constant for Steam Reforming of Methane<br>25. (a) $K_p = (p_{\text{CO}} \times (p_{\text{H}_2})^3) / (p_{\text{CH}_4} \times p_{\text{H}_2\text{O}})$ (1).<br>25. (b) Endothermic reaction ( $\Delta H > 0$ ) $\Rightarrow$ high temperature shifts equilibrium right to absorb heat, increasing Kp (2). 2 gas moles form 4 gas moles $\Rightarrow$ low pressure shifts equilibrium right towards more gas moles (2).                          | <b>[5]</b> |
| <b>Q26</b> | Q26: 26. A* Challenge: Multi-Step ICE Table with Partial Pressures<br>26. (a) $A_{\text{eq}} = 1.0$ , $B_{\text{eq}} = 1.0$ , $C_{\text{eq}} = 1.0$ , $D_{\text{eq}} = 1.0 \text{ mol}$ . Total moles = 4.0 mol (3).<br>26. (b) Mole fractions all 0.25 $\Rightarrow p_A = p_B = p_C = p_D = 2.5 \text{ atm}$ . $K_p = (2.5 \times 2.5) / (2.5 \times (2.5)^2) = 1 / 2.5 = 0.40 \text{ atm}^{-1}$ (2).                                                               | <b>[5]</b> |

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| <b>Q27</b> | Q27: 27. A* Challenge: Thermodynamic Proof of Le Chatelier's Principle<br>27. (a) $\ln K = \Delta S_{\text{system}} / R - \Delta H / (RT)$ (2).<br>27. (b) Slope of $\ln K$ vs $1/T$ is $-\Delta H / R$ . For exothermic reaction ( $\Delta H < 0$ ), slope is positive. As $T$ increases, $1/T$ decreases, so $\ln K$ decreases, meaning $K$ decreases (3).                                                                                                                                             | <b>[5]</b> |
| <b>Q28</b> | Q28: 28. A* Challenge: Calculation of Equilibrium Composition from $K_c$<br>28. (a) Initial $[N_2O_4] = 0.50$ M. Let $x = \text{mol dm}^{-3} N_2O_4$ reacted $\Rightarrow [N_2O_4]_{\text{eq}} = 0.50 - x$ , $[NO_2]_{\text{eq}} = 2x$ (1). $K_c = (2x)^2 / (0.50 - x) = 0.36 \Rightarrow 4x^2 + 0.36x - 0.18 = 0$ (2). $x = (-0.36 + \sqrt{0.1296 + 2.88}) / 8 = (-0.36 + 1.7348) / 8 = 0.172$ M (1). $[N_2O_4]_{\text{eq}} = 0.328$ M, $[NO_2]_{\text{eq}} = 0.344$ M (1).                             | <b>[5]</b> |
| <b>Q29</b> | Q29: 29. A* Challenge: Simultaneous Chemical Equilibria Analysis<br>29. (a) $K_3 = K_1 \times K_2 = 100 \times 50 = 5000$ (2).<br>29. (b) $[B] = 100[A]$ , $[C] = 50[B] = 5000[A]$ . $[A] + [B] + [C] = 5001[A] = 1.0$ M $\Rightarrow [A] = 2.0 \times 10^{-4}$ M, $[B] = 0.020$ M, $[C] = 0.980$ M (3).                                                                                                                                                                                                 | <b>[5]</b> |
| <b>Q30</b> | Q30: 30. A* Challenge: Total Pressure Effect on Dissociation Fraction<br>30. (a) Moles at eq: $A = 1 - \alpha$ , $B = 2\alpha$ . Total moles = $1 + \alpha$ (1). Partial pressures: $p_A = ((1 - \alpha) / (1 + \alpha))P$ , $p_B = ((2\alpha) / (1 + \alpha))P$ (2). $K_p = (p_B)^2 / p_A = [4\alpha^2 / (1 - \alpha^2)] P$ (1). Rearranging gives $\alpha^2 = K_p / (K_p + 4P) \Rightarrow \alpha = \sqrt{K_p / (K_p + 4P)}$ (1).                                                                      | <b>[5]</b> |
| <b>Q31</b> | Q31: 31. A* Challenge: Temperature Dependence of $K_p$ and Enthalpy Profile<br>31. (a) $\ln(0.144 / 6.82) = \ln(0.02111) = -3.858$ (1). $(1/373 - 1/298) = -6.749 \times 10^{-4} \text{ K}^{-1}$ (1). $\Delta H / R = -3.858 / (-6.749 \times 10^{-4}) = 5716 \text{ K}$ (1). $\Delta H = -5716 \times 8.31 = -47.5 \text{ kJ mol}^{-1}$ (1).<br>31. (b) Exothermic ( $\Delta H < 0$ ) (1).                                                                                                              | <b>[5]</b> |
| <b>Q32</b> | Q32: 32. A* Challenge: Equilibrium Constant Units in Terms of SI Units<br>32. (a) $K_p(\text{Pa}^{-2}) = (3.56 \times 10^{-5}) / (101325)^2 = 3.47 \times 10^{-15} \text{ Pa}^{-2}$ (2).<br>32. (b) $\text{Pa} = \text{N m}^{-2} = (\text{kg m s}^{-2}) \text{ m}^{-2} = \text{kg m}^{-1} \text{ s}^{-2} \Rightarrow \text{Pa}^{-2} = \text{kg}^{-2} \text{ m}^2 \text{ s}^4$ (2).                                                                                                                       | <b>[4]</b> |
| <b>Q33</b> | Q33: 33. A* Challenge: Gas Phase Decomposition of HI Kinetic vs Equilibrium Control<br>33. (a) $K_c = (x/2)^2 / (1 - x)^2 = 0.020 \Rightarrow x / (2(1 - x)) = \sqrt{0.020} = 0.1414 \Rightarrow x = 0.283 - 0.283x \Rightarrow 1.283x = 0.283 \Rightarrow x = 0.221$ (22.1% decomposed) (3).<br>33. (b) High temp increases rate by providing $E > E_a$ for collisions (1); endothermic reaction shifts right, increasing $K_c$ and yield (1).                                                          | <b>[5]</b> |
| <b>Q34</b> | Q34: 34. A* Challenge: Water Gas Shift Reaction Equilibrium<br>34. (a) $K_c = x^2 / ((1 - x)(2 - x)) = 1.00 \Rightarrow x^2 = 2 - 3x + x^2 \Rightarrow 3x = 2 \Rightarrow x = 0.67$ mol (3).<br>$\text{CO} = 0.33$ , $\text{H}_2\text{O} = 1.33$ , $\text{CO}_2 = 0.67$ , $\text{H}_2 = 0.67$ mol.<br>34. (b) 2 moles of gas reactants form 2 moles of gas products, so pressure changes affect numerator and denominator equally (2).                                                                   | <b>[5]</b> |
| <b>Q35</b> | Q35: 35. A* Challenge: Relate $K_p$ to Gibbs Free Energy Change $\Delta G^\circ$<br>35. (a) $\Delta G^\circ = -8.31 \times 298 \times \ln(0.141) = -2476.38 \times (-1.959) = +4851 \text{ J mol}^{-1} = +4.85 \text{ kJ mol}^{-1}$ (3).<br>35. (b) Positive $\Delta G^\circ$ means reaction is non-spontaneous under standard conditions ( $p = 1$ atm for all gases); equilibrium favors reactants (2).                                                                                                | <b>[5]</b> |
| <b>Q36</b> | Q36: 36. A* Challenge: Dissociation of Dinitrogen Tetroxide at Variable Volumes<br>36. (a) At 10 dm <sup>3</sup> : $[N_2O_4]_0 = 0.10$ M $\Rightarrow 4x^2 / (0.10 - x) = 0.0046 \Rightarrow x = 0.0102$ M ( $\alpha = 10.2\%$ ) (2). At 100 dm <sup>3</sup> : $[N_2O_4]_0 = 0.010$ M $\Rightarrow 4x^2 / (0.010 - x) = 0.0046 \Rightarrow x = 0.0028$ M ( $\alpha = 28.0\%$ ) (2).<br>36. (b) Increasing volume (dilution) increases degree of dissociation for reactions where gas moles increase (1). | <b>[5]</b> |
| <b>Q37</b> | Q37: 37. A* Challenge: Synthesis of Phosgene $\text{COCl}_2$ Equilibrium $K_p$<br>37. (a) $K_p = p_{\text{COCl}_2} / (p_{\text{CO}})^2 = 15.0 \Rightarrow 3.00 / (p_{\text{CO}})^2 = 15.0 \Rightarrow (p_{\text{CO}})^2 = 0.20 \Rightarrow p_{\text{CO}} = 0.447$ atm (3).<br>37. (b) Total pressure = $p_{\text{COCl}_2} + p_{\text{CO}} + p_{\text{Cl}_2} = 3.00 + 0.447 + 0.447 = 3.89$ atm (2).                                                                                                      | <b>[5]</b> |
| <b>Q38</b> | Q38: 38. A* Challenge: Homogeneous Gas Phase Equilibrium Calculation for $\text{SO}_3$ Formation<br>38. (a) $[\text{SO}_2] = 0.20$ M, $[\text{O}_2] = 0.10$ M. $K_c = [\text{SO}_3]^2 / ((0.20)^2(0.10)) = 280 \Rightarrow [\text{SO}_3]^2 = 280 \times 0.0040 = 1.12 \Rightarrow [\text{SO}_3] = 1.058 \text{ mol dm}^{-3}$ (3).<br>38. (b) Moles $\text{SO}_3 = 1.058 \times 2.00 = 2.12$ mol (2).                                                                                                     | <b>[5]</b> |

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| <b>Q39</b> | Q39: 39. A* Challenge: Ammonia Decomposition Kp and Partial Pressure Ratios<br>39. (a) At eq: moles NH <sub>3</sub> = 2(1-α), N <sub>2</sub> = α, H <sub>2</sub> = 3α. Total = 2 + 2α (1). K <sub>p</sub> = (p <sub>N<sub>2</sub></sub> × (p <sub>H<sub>2</sub></sub> ) <sup>3</sup> ) / (p <sub>NH<sub>3</sub></sub> ) <sup>2</sup> (2).<br>39. (b) α = 0.20. Moles: NH <sub>3</sub> = 1.6, N <sub>2</sub> = 0.2, H <sub>2</sub> = 0.6. Total = 2.4. Mole fracs: NH <sub>3</sub> =0.667, N <sub>2</sub> =0.0833, H <sub>2</sub> =0.25. Partial pressures: NH <sub>3</sub> =6.67, N <sub>2</sub> =0.833, H <sub>2</sub> =2.50. K <sub>p</sub> = (0.833 × 15.625) / 44.49 = 0.292 atm <sup>2</sup> (2). | <b>[5]</b> |
| <b>Q40</b> | Q40: 40. A* Challenge: Le Chatelier Pressure Paradox in Gas Phase Reactions<br>40. (a) Compression (decreasing volume V) immediately increases concentrations of all gases (Boyle's law) (2). System responds by shifting left (towards fewer gas moles) to reduce total moles (2). Net effect: mole fraction x <sub>NO<sub>2</sub></sub> decreases, but because volume decreased by a larger factor, equilibrium [NO <sub>2</sub> ] is higher than before compression (1).                                                                                                                                                                                                                            | <b>[5]</b> |
| <b>Q41</b> | Q41: 41. A* Challenge: Calculation of K <sub>c</sub> for Ester Hydrolysis with Added Acid Catalyst<br>41. (a) HCl catalyst is aqueous, adding initial water moles (2).<br>41. (b) Ester reacted = 0.60 mol. Acid formed = 0.60 mol, alcohol formed = 0.60 mol. Water eq = 1.00 - 0.60 = 0.40 mol. K <sub>c</sub> = (0.60 × 0.60) / (0.40 × 0.40) = 0.36 / 0.16 = 2.25 (3).                                                                                                                                                                                                                                                                                                                             | <b>[5]</b> |
| <b>Q42</b> | Q42: 42. A* Challenge: Equilibrium Constant K <sub>p</sub> of Steam Methane Reforming<br>42. (a) p <sub>CO</sub> + p <sub>H<sub>2</sub></sub> = 5.00 - 0.50 - 0.50 = 4.00 atm (1). 4 p <sub>CO</sub> = 4.00 ⇒ p <sub>CO</sub> = 1.00 atm, p <sub>H<sub>2</sub></sub> = 3.00 atm (1).<br>42. (b) K <sub>p</sub> = (1.00 × (3.00) <sup>3</sup> ) / (0.50 × 0.50) = 27.0 / 0.25 = 108 atm <sup>2</sup> (3).                                                                                                                                                                                                                                                                                               | <b>[5]</b> |
| <b>Q43</b> | Q43: 43. A* Challenge: Temperature Dependence of Solvation Equilibria<br>43. (a) Dissolution of O <sub>2</sub> gas is exothermic (ΔH < 0) (1). Increasing temperature shifts equilibrium left (towards O <sub>2</sub> gas) to absorb heat, decreasing Henry's law solubility constant K (2).<br>43. (b) Warm industrial wastewater discharges reduce dissolved oxygen levels, causing asphyxiation of fish and aerobic organisms (2).                                                                                                                                                                                                                                                                  | <b>[5]</b> |
| <b>Q44</b> | Q44: 44. A* Challenge: Multi-Gas Equilibrium Partitioning in Closed Container<br>44. (a) K <sub>c</sub> = (2x) <sup>2</sup> / ((0.80-x)(0.20-x)) ~ 4x <sup>2</sup> / (0.80 × 0.20) = 4x <sup>2</sup> / 0.16 = 1.0 × 10 <sup>-5</sup> (1). 4x <sup>2</sup> = 1.6 × 10 <sup>-6</sup> ⇒ x <sup>2</sup> = 4.0 × 10 <sup>-7</sup> ⇒ x = 6.32 × 10 <sup>-4</sup> M (1). [NO] = 2x = 1.26 × 10 <sup>-3</sup> mol dm <sup>-3</sup> (1).<br>44. (b) K <sub>c</sub> is extremely small (10 <sup>-5</sup> ), so x << 0.20 M (approximation error < 1%) (2).                                                                                                                                                       | <b>[5]</b> |
| <b>Q45</b> | Q45: 45. A* Challenge: Van 't Hoff Isochore Derivation & Graph Analysis<br>45. (a) Gradient = -ΔH / R = +4500 K ⇒ ΔH = -4500 × 8.31 = -37395 J mol <sup>-1</sup> = -37.4 kJ mol <sup>-1</sup> (exothermic) (2).<br>45. (b) Y-intercept = ΔS <sub>sys</sub> / R = -12.5 ⇒ ΔS <sub>sys</sub> = -12.5 × 8.31 = -103.9 J K <sup>-1</sup> mol <sup>-1</sup> (3).                                                                                                                                                                                                                                                                                                                                            | <b>[5]</b> |
| <b>Q46</b> | Q46: 46. A* Challenge: Gas Phase Equilibrium Dissociation of N <sub>2</sub> O <sub>4</sub> at High Altitude<br>46. (a) α = 0.20 at 1.0 atm ⇒ K <sub>p</sub> = 4(0.20) <sup>2</sup> / (1 - (0.20) <sup>2</sup> ) × 1.0 = 0.16 / 0.96 = 0.167 atm (2).<br>46. (b) 0.167 = [4 α <sup>2</sup> / (1 - α <sup>2</sup> )] × 0.50 ⇒ 4 α <sup>2</sup> / (1 - α <sup>2</sup> ) = 0.3333 ⇒ 4.3333 α <sup>2</sup> = 0.3333 ⇒ α <sup>2</sup> = 0.0769 ⇒ α = 0.277 (27.7% dissociated) (3).                                                                                                                                                                                                                          | <b>[5]</b> |
| <b>Q47</b> | Q47: 47. A* Challenge: Industrial Synthesis of Hydrogen Cyanide Equilibrium<br>47. (a) Exothermic reaction favoured by low T, but 1100 °C required for high rate (1). 3.5 gas moles → 4 gas moles favoured by low pressure (2).<br>47. (b) Extremely high activation energy requires Pt-Rh catalyst at 1100 °C for millisecond contact time (2).                                                                                                                                                                                                                                                                                                                                                       | <b>[5]</b> |
| <b>Q48</b> | Q48: 48. A* Challenge: Equilibrium Constant of Carbonate Buffer System<br>48. (a) Higher p <sub>CO<sub>2</sub></sub> increases [CO <sub>2</sub> (aq)], shifting equilibrium right to produce more H <sup>+</sup> ions, lowering ocean pH (acidification) (3).<br>48. (b) Increased H <sup>+</sup> reacts with CaCO <sub>3</sub> marine shells/coral, dissolving them (2).                                                                                                                                                                                                                                                                                                                              | <b>[5]</b> |
| <b>Q49</b> | Q49: 49. A* Challenge: Equilibrium Analysis of Dehydration of Hydrated Cobalt Chloride<br>49. (a) In dry air (low p <sub>H<sub>2</sub>O</sub> ), equilibrium shifts right (towards blue CoCl <sub>2</sub> ) to replenish H <sub>2</sub> O gas (1.5). In humid air (high p <sub>H<sub>2</sub>O</sub> ), equilibrium shifts left (towards pink hydrate) (1.5).<br>49. (b) K <sub>p</sub> = (p <sub>H<sub>2</sub>O</sub> ) <sup>6</sup> (2).                                                                                                                                                                                                                                                              | <b>[5]</b> |

|            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |            |
|------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|
| <b>Q50</b> | Q50: 50. A* Challenge: Complete Equilibria & Thermodynamics Master Synthesis<br>50. (a) $\Delta S_{\text{surr}} = -(-108000)/298 = +362.4 \text{ J K}^{-1} \text{ mol}^{-1}$ . $\Delta S_{\text{tot}} = -132 + 362.4 = +230.4 \text{ J K}^{-1} \text{ mol}^{-1}$ (2). $\ln K_p = 230.4 / 8.31 = 27.72 \Rightarrow K_p = e^{27.72} = 1.09 \times 10^{12} \text{ atm}^{-1}$ (2).<br>50. (b) $T_{\text{boundary}} = \Delta H / \Delta S_{\text{sys}} = 108000 / 132 = 818.2 \text{ K}$ (545.0 °C) (2). | <b>[6]</b> |
|------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|

## SECTION D: EMPIRICAL EXAMINER FAQs &amp; COMMON TRAPS (10 FAQs) — 13A

**FAQ #1: Omitting Solids/Liquids from  $K_c$  and  $K_p$** Category: Heterogeneous  
Equilibria**COMMON EXAMINER TRAP / MISTAKE:**

Including pure solids or liquids in  $K_c/K_p$  expressions.

**MODEL ANSWER / EXAMINER EXPECTATION:**

Concentrations of pure solids and pure liquid solvents are constant and built into the equilibrium constant value. Omit them entirely from  $K_c$  and  $K_p$ .

**FAQ #2: Units of  $K_c$  and  $K_p$  Deduction**

Category: Units Calculation

**COMMON EXAMINER TRAP / MISTAKE:**

Writing dimensionless or standard units without checking stoichiometry.

**MODEL ANSWER / EXAMINER EXPECTATION:**

Units depend on mole change: substitute ( $\text{mol dm}^{-3}$ ) into  $K_c$  or ( $\text{atm/kPa}$ ) into  $K_p$ . If gas moles on left equal gas moles on right,  $K$  is dimensionless.

**FAQ #3: Effect of Pressure on  $K_p$  Value**

Category: Pressure Paradox

**COMMON EXAMINER TRAP / MISTAKE:**

Claiming increasing total pressure increases the value of  $K_p$ .

**MODEL ANSWER / EXAMINER EXPECTATION:**

Total pressure changes shift the EQUILIBRIUM POSITION, but  $K_p$  itself is CONSTANT at a fixed temperature. The partial pressures adjust to maintain exact  $K_p$ .

**FAQ #4: Effect of Temperature on  $K$  Sign**

Category: Temperature Effect

**COMMON EXAMINER TRAP / MISTAKE:**

Confusing direction of  $K$  shift for exothermic vs endothermic reactions.

**MODEL ANSWER / EXAMINER EXPECTATION:**

For EXOTHERMIC ( $\Delta H < 0$ ), heating DECREASES  $K$ . For ENDOTHERMIC ( $\Delta H > 0$ ), heating INCREASES  $K$ . (Remember:  $K$  changes ONLY with temperature).

**FAQ #5: Catalyst Effect on Equilibrium Yield**

Category: Catalyst Fallback

**COMMON EXAMINER TRAP / MISTAKE:**

Stating a catalyst increases equilibrium yield or shifts position.

**MODEL ANSWER / EXAMINER EXPECTATION:**

A catalyst speeds up forward and reverse reactions by the EXACT SAME FACTOR. It reaches equilibrium faster but has ZERO effect on yield or  $K$ .

**FAQ #6: Mole Fraction vs Partial Pressure**Category: Partial Pressure  
Calculation**COMMON EXAMINER TRAP / MISTAKE:**

Using mole fraction directly in  $K_p$  instead of partial pressure.

**MODEL ANSWER / EXAMINER EXPECTATION:**

Partial pressure  $p_A$  = mole fraction  $x_A$  x TOTAL PRESSURE. You must multiply by total pressure before inserting into  $K_p$ .

**FAQ #7: ICE Table Initial Moles vs Equilibrium Moles****Category: ICE Table Trap****COMMON EXAMINER TRAP / MISTAKE:**

Inserting initial moles directly into  $K_c$  expression without subtracting change  $x$ .

**MODEL ANSWER / EXAMINER EXPECTATION:**

Always construct an ICE table (Initial, Change, Equilibrium). Moles at equilibrium = Initial +/– stoichiometric change  $x$ , converted to concentration ( $[ ] = \text{mol} / V$ ).

**FAQ #8: Relating Total Entropy to K Formula****Category: Thermodynamic Formula****COMMON EXAMINER TRAP / MISTAKE:**

Using  $\log_{10}$  instead of natural log  $\ln$  in  $\Delta S_{\text{total}} = R \ln K$ .

**MODEL ANSWER / EXAMINER EXPECTATION:**

The formula uses natural logarithm ( $\ln$  / base  $e$ ).  $\ln K = \Delta S_{\text{total}} / R \Rightarrow K = e^{(\Delta S_{\text{total}} / R)}$ .  
 $R = 8.31 \text{ J K}^{-1} \text{ mol}^{-1}$ .

**FAQ #9: Volume Cancellation in  $K_c$  Expressions****Category: Volume Shortcut****COMMON EXAMINER TRAP / MISTAKE:**

Dividing by volume  $V$  when gas moles on both sides are equal.

**MODEL ANSWER / EXAMINER EXPECTATION:**

When sum of moles of reactants equals sum of moles of products (e.g. esterification), volume  $V$  cancels out completely.  $K_c$  can be calculated directly from equilibrium moles.

**FAQ #10: Reaction Quotient  $Q$  vs Equilibrium Constant  $K$** **Category: Direction of Shift****COMMON EXAMINER TRAP / MISTAKE:**

Confusing direction of shift when  $Q > K$ .

**MODEL ANSWER / EXAMINER EXPECTATION:**

If  $Q < K$ , system shifts RIGHT to form more products. If  $Q > K$ , system shifts LEFT to form more reactants until  $Q = K$ .